

Problem-Solution Fit

Date	1 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

This canvas explains how the Voice-Based Concept Understanding Analyzer addresses the needs of students by identifying learning problems, understanding their causes, and providing an AI-powered voice-based solution for concept evaluation.

1. Customer Segment (CS) Students, Teachers, Trainers and Educational Institutions.	6. Customer Limitations (CL) Limited internet access, low-end devices, lack of personalized feedback.	5. Available Solutions (AS) Online quizzes, LMS platforms and manual evaluation with limited feedback.
2. Problems / Pains (PR) Difficulty in measuring conceptual understanding through traditional exams.	9. Problem Root Cause (RC) Existing systems evaluate only written answers and ignore verbal understanding.	7. Behavior (BE) Students frequently depend on rote learning instead of concept explanation.
3. Triggers to Act (TR) Need for continuous assessment and AI-assisted learning. 4. Emotions (EM) Before: Anxiety and confusion. After: Confidence and improved understanding.	10. Your Solution (SL) An AI-powered application that accepts voice explanations, transcribes speech, evaluates semantic similarity, analyzes audio features, calculates scores, and generates detailed PDF reports.	8. Channels of Behavior (CH) Online: Web Application, LMS, Cloud Platform. Offline: Classroom Demonstrations and Local Usage.